ProCV Resume Builder

A CV Creation and Management System in x86 Assembly

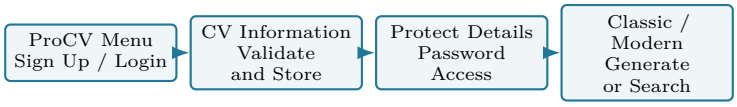
CSE341 Project Report

1. Introduction

ProCV Resume Builder is a text-based CV creation and management application developed using **x86 Assembly Language**. The system guides users from profile login through personal information, education, work experience, and skills entry to generation of a formatted CV. It also incorporates password-based protection for sensitive information and a search function for retrieving stored CV records.

2. System Workflow

The application uses a menu-driven workflow to simplify navigation. Following login, users can complete or update CV sections, select a presentation template, and search for existing entries.



3. Implemented Functionality

Feature	Status	Functionality in the final project
User-friendly interface	Implemented	A labeled *ProCV Resume Builder* welcome screen and clear text-based menus guide users through available options.
Profile login	Implemented	Users enter their ID and password to access their profile; the password also enables decryption of protected information.
Personal information	Implemented	Collects name, age, contact number, email, and address, with basic input-validation checks for age and contact details.
Education	Implemented	Supports multiple qualifications, recording degree, institution, and graduation year for each entry.
Work experience	Implemented	Supports multiple employment entries with job title, company name, and duration.
Skills	Implemented	Records up to eight skills, each assigned a proficiency level: Beginner, Intermediate, or Expert.
CV templates	Implemented	Generates CV output using either the Classic or the Modern formatting style selected by the user.
Encryption/decryption	Implemented	Applies basic encryption to sensitive fields, including email and contact number, and decrypts them using the user's password.
Search and retrieval	Implemented	Searches stored CV entries by identifying information, such as name or email, and displays the matching record.

4. Technical Implementation

The application uses an x86 Assembly menu system with dedicated procedures for authentication, personal details, qualifications, work history, skills, CV generation, and search. DOS interrupts handle console input/output. Fixed-size data buffers, indexed records, loops, comparisons, and conditional jumps support user interaction, data storage, and field validation.

The login password is also used to decrypt encrypted email addresses and contact numbers when authorized access is required. The CV generator assembles the saved sections into either a **Classic** or **Modern** text-based layout, while the search module locates entries by name or email and displays the matching CV.

5. Functional Verification

The key checks include menu navigation, login, valid and invalid profile inputs, entry of multiple education and work records, all eight skill slots, both CV styles, password-based decryption, and retrieval by name or email. Each test exercises a distinct part of the final project's nine implemented features.

6. Conclusion

ProCV Resume Builder demonstrates a complete console-based CV management workflow in x86 Assembly Language. By combining structured data entry, validation, flexible CV layouts, sensitive-data encryption and decryption, and searchable records, the system provides a practical way to prepare and manage professional CVs.